# CY4001: Industrial Chemistry

## Course assessment and Evaluation scheme

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|---|---|
| Course Code and Title | **CY4001: Industrial Chemistry** |
| Course credit | **3 Credit** |
| Course category | **B. Tech. (SME Courses; General Category Elective)** |
| Prerequisite, if any | **CY1040 Basic Chemistry for Engineers** |
| **Evaluation scheme** | **Test-1 = 25, Test-2 = 25, Assignment=10 and End Sem = 40 Marks** |
| Course Instructor | **Dr. Umadevi D (Chemistry department)**<br>**Email: umadevi@iitpkd.ac.in**<br>**Phone: 9790362235** |

# CY4003: Introduction to Nanoscience and Nanotechnology

## Course Objectives

*This course aims to equip students with an understanding of the chemistry behind real-world industries, including industrial processes, materials, energy systems, safety, and sustainability.*

**O1** : To help students discover how chemistry drives real-world industries

**O2** : To expose students to modern industrial materials and technologies,

**O3** : To build awareness of industrial safety and hazard management

**O4** : To introduce practical chemistry-based approaches for clean water, waste management, and pollution control

**O5** : To inspire sustainable thinking through green chemistry and energy-efficient technologies

# CY4003: Introduction to Nanoscience and Nanotechnology

## Learning Outcomes

- **See how chemistry works behind real industries,** rom fuels and materials to large-scale manufacturing that supports everyday life.
- **Understand modern industrial materials and technologies,** including polymers, nanomaterials, and energy-related systems used in engineering applications.
- **Appreciate the importance of industrial safety** and learn how engineers identify hazards and prevent accidents in real industrial environments.
- **Relate classroom concepts to real-world challenges**, such as clean energy, clean water, and sustainable industrial development.

*This course is not about becoming a chemist. It's about seeing chemistry at work in the real world.*

# CY4003: Introduction to Nanoscience and Nanotechnology

## Learning Through Innovation

To enhance student engagement and deepen conceptual understanding, this course adopts a range of innovative teaching approaches.

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|---|---|---|
| **Audio-Visual Learning** | : | To support diverse learning styles and enhance understanding of abstract concepts |
| **Case studies** | : | To connect theoretical concepts with real-world applications and contemporary challenges |
| **Peer instruction techniques** | : | To promote conceptual understanding through think-vote-justify cycles and guided peer discussion |
| **Flipped classroom sessions** | : | To facilitate pre-class concept familiarization and enable active, discussion-based learning during class time |
| **Narrative pedagogy** | : | To contextualize scientific concepts, humanize science, and motivate student engagement |

# CY4003: Introduction to Nanoscience and Nanotechnology

## Is This Course for Me?

- CY4001 is not “just a chemistry class” - it’s a “chemistry-centered” class that shows how chemistry operates behind real-world industries, technologies, and everyday products.

- This course offers a fresh perspective on familiar topics, such as fuels, materials, water, and energy, by connecting basic chemistry with industrial practice, safety, and sustainability..

- It provides a strong foundation to understand industrial materials and processes, helping you appreciate how chemistry supports modern technologies, emerging materials, and sustainable development across disciplines.

***If you’ve ever wondered how everyday materials become real products, this course answers that question***

# CY4003: Introduction to Nanoscience and Nanotechnology

## Time and Venue

**Slot : G**-slot

**Time:** Tuesday 10.30 to 11.45 am
Thursday 9.00 to 10.15 am

**Venue: A01-007** (Lecture Theatre) Saraswati block

- Starts five minutes late and end 5 minutes earlier
- Attendance – as per institute policy (> 85%)
- All the class information will be shared in the Whatsapp group.

# Evaluation scheme

| Date | Assessment | Marks |
|---|---|---|
| Feb., 2026 | Test-1 | 25 |
| Mar., 2026 | Test-2 | 25 |
| Apr., 2026 | Assignment | 10 |
| Apr., 2026 | End Sem | 40 |
| | Total | 100 |

Marks will be awarded on an absolute scale, but final grades will be assigned based on relative performance

*Last Instructional Day for CY4001 - April 23, 2026*

# CY4003: Introduction to Nanoscience and Nanotechnology

## Creating a Positive Learning Space

- ❑ 🪑 **Fill the front rows first**. It helps everyone stay more engaged.
- ❑ 🥤 **Water and light snacks are okay**, as long as it doesn't distract others or leave a mess.
- ❑ ❓ **Don't hesitate to ask questions**. No question is too small or silly. We're here to learn!
- ❑ 🤝 **Respect everyone's opinion**. Healthy discussion is welcome, but let's be kind and listen too.
- ❑ 🤫 **Let's keep our voices a little low so everyone can follow**

Let's learn, grow, and make it enjoyable for all. Your cooperation makes a big difference!

# CY4003: Introduction to Nanoscience and Nanotechnology

## Recipes for success in CY4003

Want to do well in this course? Here's your secret recipe!

- ❑ 💁♀️ **Attend and really listen** during lectures, it is where most concepts become clear!
- ❑ 📚 **Read the materials provided** – they are crafted to make your learning easier.
- ❑ 💬 **Join classroom discussions** – you will learn a lot from your peers (and its fun too!).
- ❑ 🙋 **Ask for help when needed** – no shame in asking! Everyone needs a hand sometimes.

Consistency beats intensity. Just show up, stay curious, and keep moving forward!

# CY4001: Industrial Chemistry

## Course plan and organization

## Test-1: 25 marks

### Unit 1: Introduction to Chemical Industry

Role of chemistry in industry, Chemistry in process and manufacturing industry, Development and characteristics of the chemical Industry. Role of chemistry in scale up procedure and factors involve in moving from bench to manufacturing plant

### Unit 2: Industrial Safety & Hazards Management

Introduction, Toxicology, Industrial Hygiene, Basics of Fires and Explosion, Hazard identification, Risk Assessment, Accident Investigations and Case Histories.

### Unit 3: Fuel Chemistry

Review of energy sources (renewable and non-renewable). Classification of fuels (a) Coal: Classification - Pulverized Coal., Selection Criteria, Proximate and Ultimate Analysis, Liquefaction of Coal (Fischer-Tropsch method); Gasification of Coal (water gas). (b) Petroleum: Classification, Types of Cracking, Fuel Rating, LPG, Natural Gas, Producer Gas, Water Gas, Bio Gas. (c) Oxygenated fuels viz Biodiesel and Ethanol: their production and characterization. Alternate Fuels

# CY4001: Industrial Chemistry

## Course plan and organization

## Test-2: 25 marks

### Unit 4: Polymer Chemistry:

Terminology, Classification polymers, Types of polymerization techniques. Preparation, properties and applications of: phenol-formaldehyde resins, synthetic rubbers, silicones, polyphosphazenes, epoxy resins. Introduction to conducting polymers (polythiophene, polyaniline) and their application in advanced technologies

### Other Chemical Industries

Paint and Coatings, Adhesives and Adhesion, Explosives, Cement, Ceramics, and Composites

## Assignment : 10 marks (Topics/problems will be given later)

# CY4001: Industrial Chemistry

## End-Sem: 40 marks

### Unit 5: Photochemistry:

Introduction, significance, fluorescence, phosphorescence, chemiluminescence and photosensitization, applications in light emitting diodes (LED) and environment related examples.

### Unit 6: Nanochemistry:

Materials, Surfaces, Introduction to nanochemistry, synthesis, properties and applications of carbon-based nanomaterials (fullerenes, carbon nanotubes, quantum dots and graphene) in electronic and nanodevices, Metal organic frameworks

### Unit 7, 8 9: Water and Its Treatment, Industrial Waste Management, Social Issues and Environment

Sources of water, impurities in water, hardness of water, its disadvantages, boiler scale and its prevention. Treatment of water - Lime soda method, zeolite process, ion-exchange process and domestic water treatment.

Types of industries and industrial waste characteristics; management strategies for pollution prevention and waste minimization; hazardous waste management.

Sustainable Development-Introduction and measures, Greenhouse Effect-Causes and impacts, Measures to check Global Warming; Acid Rain Causes, Effects and Ways to Control; Green Chemistry- carbon zero, principles and industrial examples.

# CY4001: Industrial Chemistry

## Textbooks:


1. Survey of Industrial Chemistry by P. J. Chenier (3rd Edition) , 2002, Kluwer Academic/Plenum Publishers. ISBN: 0-306-47246-5.

2. A Text Book of Engineering Chemistry: O. G. Palanna (4th reprint 2012), McGraw Hill, New Delhi. ISBN: 9780070146105

3. C. A. Heaton, Introduction to Industrial Chemistry, 3rd Ed, Springer, New York (2009). ISBN: 978-0751402728.

4. M. Farhat Ali, Handbook of Industrial Chemistry, 1st Ed, McGraw-Hill Education (2005). ISBN: 978-0071410373.


## References:


1. K. Weissermel, Industrial Organic Chemistry, 3rd Ed, Wiley-VCH, New York (2003). ISBN: 3- 527-28838-4 Gb.

2. O.V. Roussak, H. D. Gesser, Applied Chemistry: A Textbook for Engineers and TechnologistsSpringer US (2013). ISBN: 978-1-4614-4262-2.

3. E. Stocchi: Industrial Chemistry, Vol-I, Ellis Horwood Ltd. UK (1990). ISBN: 978-0134573182